

Inter-arc continuity mathematics: before and after review

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Quantities used by both implementations. For body b and constrained arc pair $k = (\ell, r)$, evaluated at connection epoch $t_{c,bk}$,

$$\begin{aligned}\mathbf{d}_{bk} &:= \mathbf{x}_{r,b}(t_{c,bk}) - \mathbf{x}_{\ell,b}(t_{c,bk}), \\ \mathbf{D}_{bk} &:= \frac{\partial \mathbf{d}_{bk}}{\partial \mathbf{p}} = \mathbf{M}_{r,b}(t_{c,bk}) - \mathbf{M}_{\ell,b}(t_{c,bk}), \\ \tilde{\mathbf{D}}_{bk} &:= \mathbf{D}_{bk} \text{diag}(\mathbf{s})^{-1}.\end{aligned}$$

Here $\mathbf{p}$ is the estimated parameter vector, $\mathbf{M}$ denotes the relevant rows of the full state-transition/sensitivity matrix, and $\mathbf{s}$ contains Tudat's observation-design-matrix column normalization factors. Let m be the number of scalar observation residuals and

$$\bar{\mathbf{C}}_{bk} := \frac{1}{2}(\mathbf{C}_{bk} + \mathbf{C}_{bk}^T), \quad m_d := \sum_{b,k} \text{rank}(\bar{\mathbf{C}}_{bk}).$$

$\mathbf{C}_{bk}$ is the *continuity* weight supplied through `weight_matrices`, or the diagonal matrix constructed from `position_weights` and `velocity_weights`. It is distinct from the observation weight matrix $\mathbf{W}_o$. The symmetric part matters only for input asymmetry within the accepted numerical tolerance.

Before the mathematical correction. The code used

$$\mathbf{W}_{bk}^- = \frac{1}{\mu_s m_d} \mathbf{C}_{bk},$$

where μ_s is the settings object's `constraint_scaling_factor`. Ignoring round-off-level asymmetry for this paragraph, the normal-equation additions were

$$\Delta \tilde{\mathbf{A}}^- = \sum_{b,k} \tilde{\mathbf{D}}_{bk}^T \mathbf{W}_{bk}^- \tilde{\mathbf{D}}_{bk}, \quad \Delta \tilde{\mathbf{q}}^- = - \sum_{b,k} \tilde{\mathbf{D}}_{bk}^T \mathbf{W}_{bk}^- \mathbf{d}_{bk}.$$

Consequently, the solve represented the local quadratic objective

$$J_{\text{solve}}^- = \frac{1}{2} \mathbf{r}^T \mathbf{W}_o \mathbf{r} + \frac{1}{2} \sum_{b,k} \mathbf{d}_{bk}^T \mathbf{W}_{bk}^- \mathbf{d}_{bk}.$$

However, iteration selection added the reported constraint cost without the factor 1/2:

$$J_{\text{select}}^- = \frac{1}{2} \mathbf{r}^T \mathbf{W}_o \mathbf{r} + \sum_{b,k} \mathbf{d}_{bk}^T \mathbf{W}_{bk}^- \mathbf{d}_{bk}.$$

Thus selection evaluated a different observation/constraint balance from the one used by the correction step. The constraint also lacked the factor m required to retain the relative normalization of the averaged observation and discontinuity terms in the cited Orbit14 model.

For an accepted slightly non-symmetric $\mathbf{C}$, the old implementation had an additional inconsistency: the explicitly symmetrized normal matrix and scalar quadratic cost used $\bar{\mathbf{C}}$, while the right-hand side used raw $\mathbf{C}$.

After the mathematical correction. Every continuity term now uses

$$\mathbf{W}_{bk}^+ = \frac{m}{\mu_s m_d} \bar{\mathbf{C}}_{bk}.$$

The single implemented objective, its linear term, and its curvature are

$$J^+ = \frac{1}{2} \mathbf{r}^T \mathbf{W}_o \mathbf{r} + \frac{1}{2} \sum_{b,k} \mathbf{d}_{bk}^T \mathbf{W}_{bk}^+ \mathbf{d}_{bk},$$

$$\Delta \tilde{\mathbf{A}}^+ = \sum_{b,k} \tilde{\mathbf{D}}_{bk}^T \mathbf{W}_{bk}^+ \tilde{\mathbf{D}}_{bk}, \quad \Delta \tilde{\mathbf{q}}^+ = - \sum_{b,k} \tilde{\mathbf{D}}_{bk}^T \mathbf{W}_{bk}^+ \mathbf{d}_{bk}.$$

With $\mathbf{d}(\mathbf{p} + \delta \mathbf{p}) \simeq \mathbf{d} + \mathbf{D} \delta \mathbf{p}$, these are exactly the Hessian/Gauss–Newton term and negative gradient contribution of the same quadratic objective. The reported constraint cost is now the second term of J^+ , so best-iteration selection and the solve use the same model.

The Orbit14 equations used as the reference are

$$Q_{\text{ref}} = \frac{1}{m} \mathbf{r}^T \mathbf{W}_o \mathbf{r} + \sum_{b,k} \frac{1}{\mu_s m_d} \mathbf{d}_{bk}^T \bar{\mathbf{C}}_{bk} \mathbf{d}_{bk}.$$

Therefore

$$J^+ = \frac{m}{2} Q_{\text{ref}},$$

which has the same minimizer as Q_{ref} and matches Tudat’s existing $\frac{1}{2} \mathbf{r}^T \mathbf{W}_o \mathbf{r}$ convention. A separate μ_s per settings object is the implementation’s direct extension of the paper’s single penalty parameter.

What did not change.

- The discrepancy remains right-arc state minus left-arc state.
- Its partial remains right-arc variational rows minus left-arc variational rows.
- The estimation right-hand-side sign remains negative for the continuity term.
- Component masks and dense positive-semidefinite weights remain supported; m_d is their total rank, generalizing $6(n-1)$.

References. G. Lari et al., “Orbit determination methods for interplanetary missions: the Orbit14 software,” *Experimental Astronomy* (2021), Eqs. (4) and (28), [doi:10.1007/s10686-021-09823-8](https://doi.org/10.1007/s10686-021-09823-8). For comparison, E. M. Alessi et al., “Orbit determination strategy and accuracy for the BepiColombo radio science experiment,” *Advances in Space Research* (2012), Eq. (5), uses a different common-denominator normalization; the implementation follows the later Orbit14 formulation above.